

Eenheid 1.10

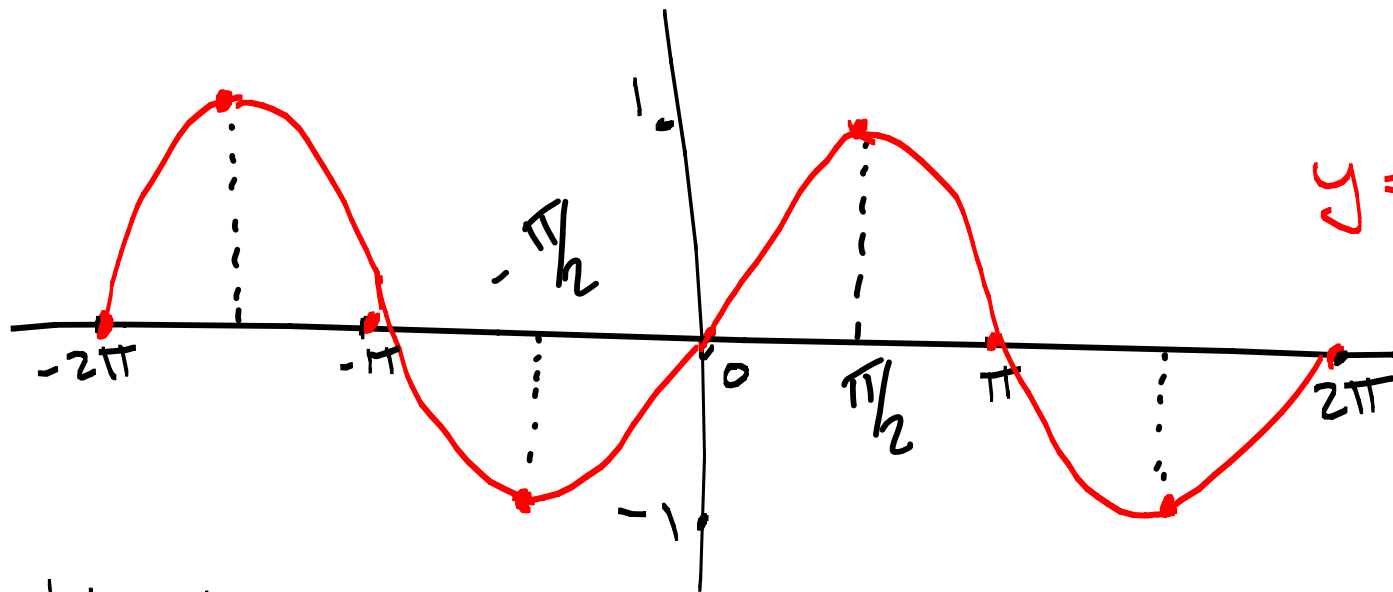
Note Title

2016/02/21

Inverse trigonometrische functies

Stewart bl. 63-66

$$\underline{y = \sin x}$$



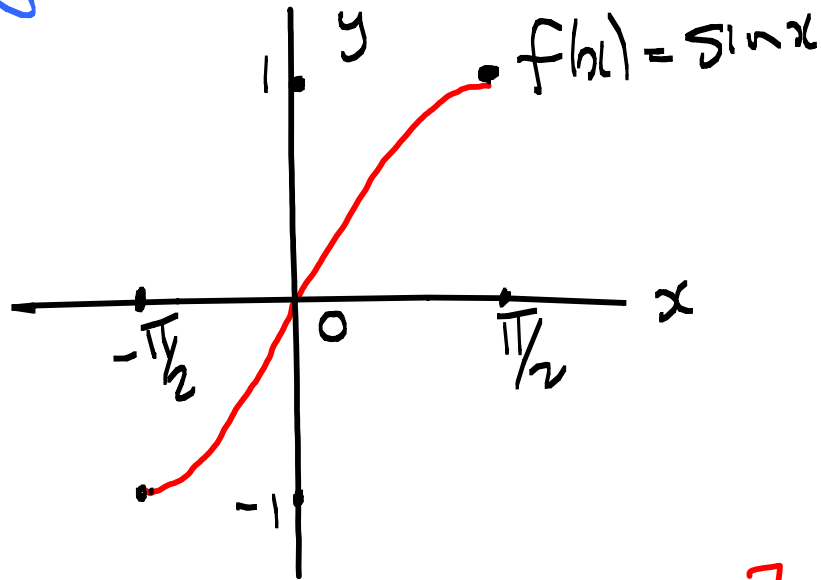
Het $y = \sin x$ is inverse functie??

Ja / Nee? Motiveer! niet - een-tot-een

Plan? Inverse??

Als

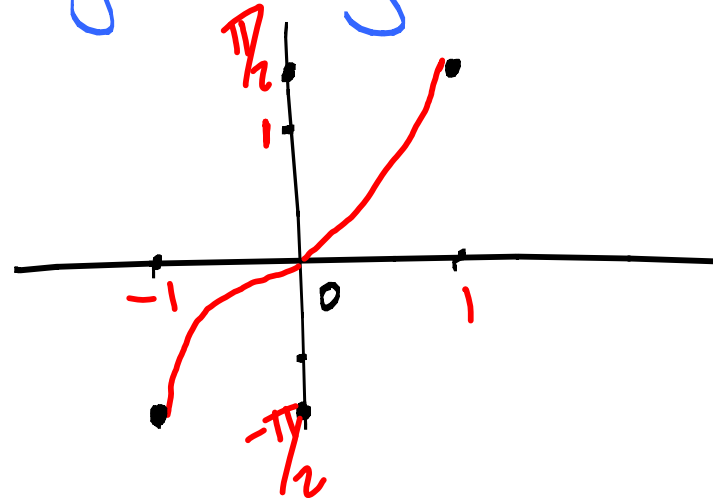
$y = \sin x$



$$f: [-\pi/2, \pi/2] \rightarrow [-1, 1]$$

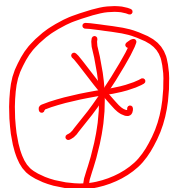
Dan

$y = \arcsin x$
 $y = \log \sin x = \sin^{-1} x$

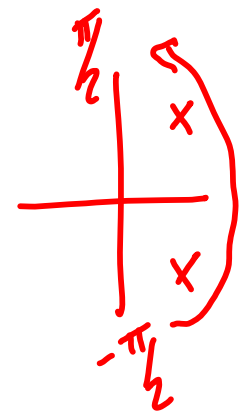


$$f^{-1}: [-1, 1] \rightarrow [-\pi/2, \pi/2]$$

$$y = \log \sin x = \arcsin x = \sin^{-1} x \neq \frac{1}{\sin x}$$



Definisie: $*y = \text{bg sin } x \Leftrightarrow \sin y = x$
 (volledig) en $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$



Kansellasie Vergelykings:

1. $\sin(\text{bg sin } x) = x$ as $x \in [-1, 1]$
2. $\text{bg sin}(\sin x) = x$ as $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Los op:

① $\sin x = -\frac{1}{2}$

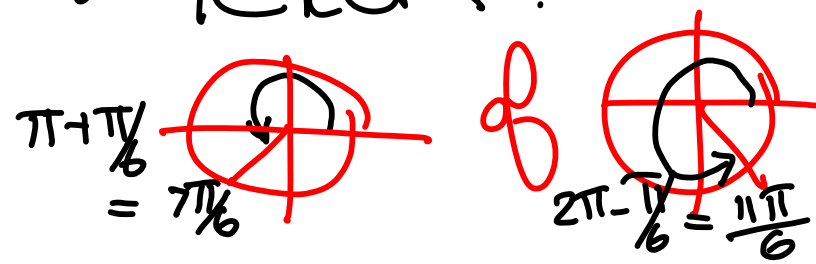
(oneindig baie antwoorde)

$x = \frac{7\pi}{6} + 2k\pi$ of $x = \frac{11\pi}{6} + 2k\pi \quad | k \in \mathbb{Z}$

(of $= -\frac{\pi}{6} + 2k\pi$)

• Verwysings: $\frac{\pi}{6}$

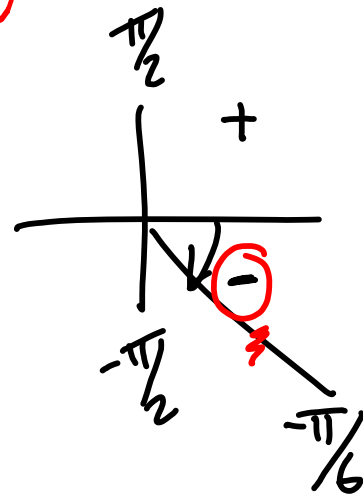
• Tekens:



② $y = \arcsin(-\frac{1}{2})$ (Gebruik definisie) (slegs een unieke antwoord)

$\Leftrightarrow \sin y = -\frac{1}{2}$ en $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\Rightarrow y = -\frac{\pi}{6}$



③ $\sin^{-1}(\sin \frac{7\pi}{3}) = y$

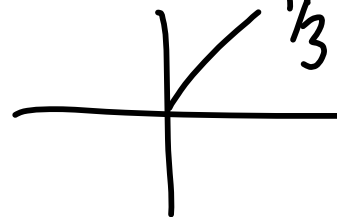
As ek kansellasie eienskap? gebruik
 ?? lyk dit of $y = \frac{7\pi}{3}$??

$\sin \frac{7\pi}{3} = \frac{\sqrt{3}}{2}$

Verwysinghoek is $\frac{\pi}{3}$

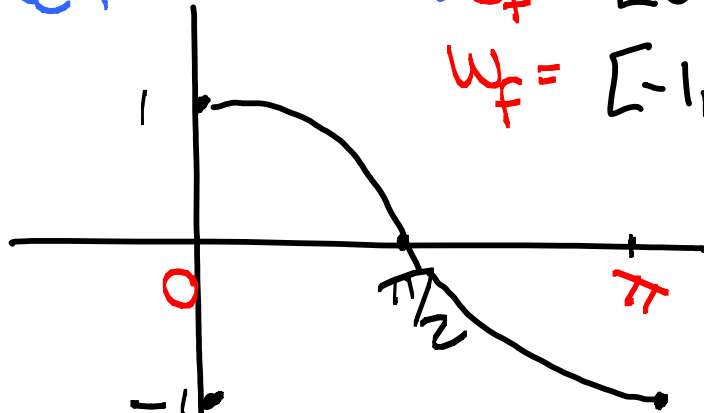
$\sin^{-1}(\frac{\sqrt{3}}{2}) = y \Leftrightarrow \sin y = \frac{\sqrt{3}}{2}$ en $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

Dus $y = \frac{\pi}{3}$



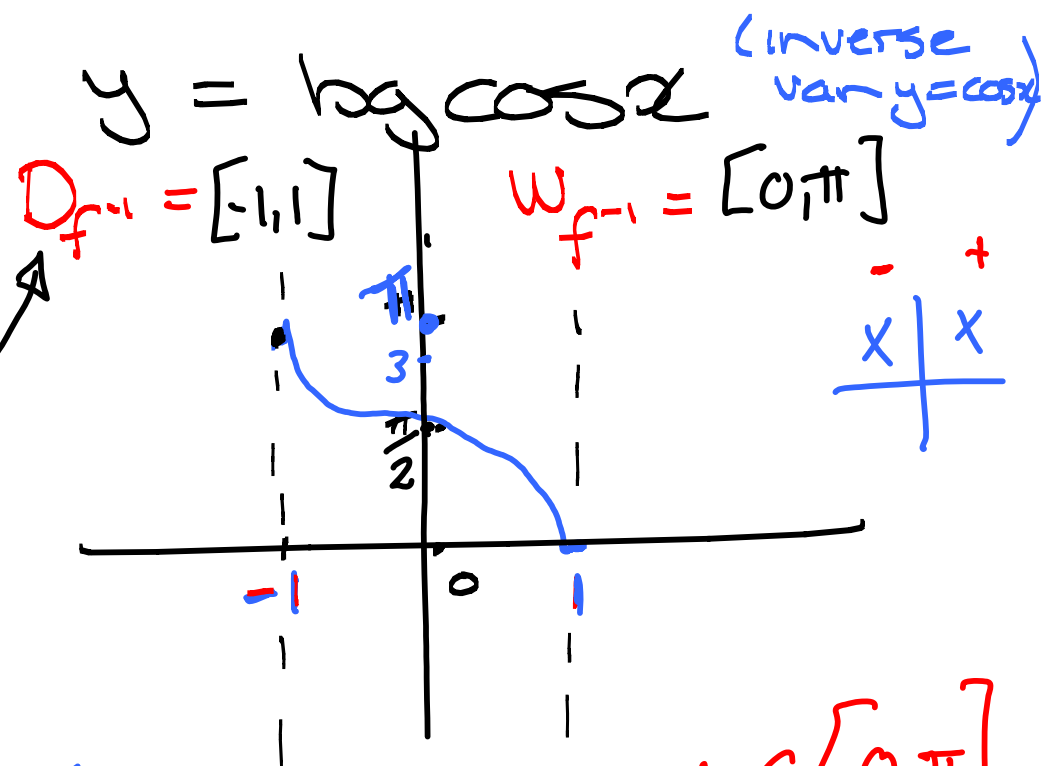
- $y = \cos x$ moet een-tot-een wees om inverse funktie te hē.

Plan:



$$D_f = [0, \pi]$$

$$W_f = [-1, 1]$$



* Definitie: $y = \cos^{-1} x \Leftrightarrow \cos y = x$ en $y \in [0, \pi]$

Kansellatie vergelykingen:

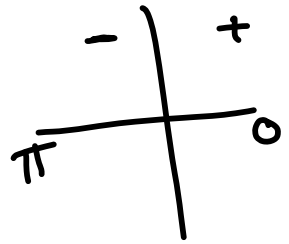
1. $\cos(\text{bg} \cos x) = x$ vir $x \in [-1, 1]$

2. $\text{bg} \cos(\cos x) = x$ vir $x \in [0, \pi]$

① Los op: $y = \arccos(\cos(0.5))$ (gebruik definitie)

$\Leftrightarrow \cos y = \frac{1}{2}$ en $y \in [0, \pi]$

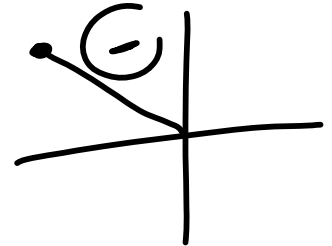
$\Rightarrow y = \frac{\pi}{3}$ (een unieke antw)



Vind: ② $y = \arccos(-\frac{\sqrt{3}}{2})$

$\Leftrightarrow \cos y = -\frac{\sqrt{3}}{2}$ en $y \in [0, \pi]$

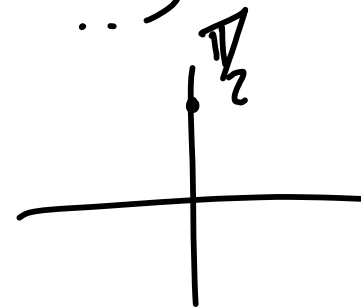
$\Rightarrow y = \frac{5\pi}{6}$ ($\pi - \frac{\pi}{6}$)



$$\textcircled{3} \quad y = \arccos(\cos(-\frac{\pi}{2})) = \arccos(\cos(0))$$

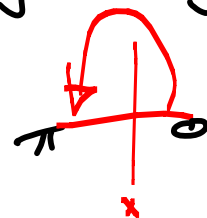
$$\Leftrightarrow \cos y = 0, \quad y \in [0, \pi]$$

$$\Rightarrow y = \frac{\pi}{2}$$



of $\arccos$ kan ook die kansellasie eienskap gebruik.

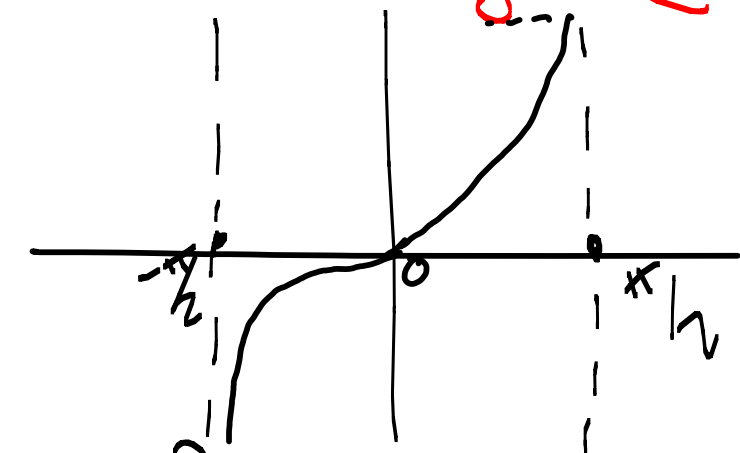
$$y = \arccos(\cos(\frac{\pi}{2}))$$



Probleem!! kan nie sê $y = -\frac{\pi}{2}$ nie
want $y \in [0, \pi]$

$$\text{Dus } y = \frac{\pi}{2}$$

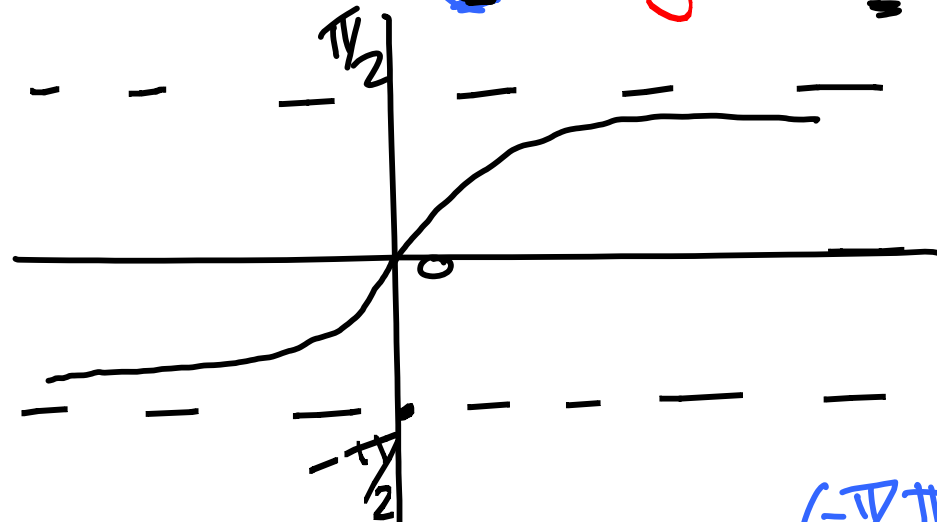
• $y = \tan x$ • een-een
duidig?
Plan: Laat $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$
 $y \in \mathbb{R}$



Definitie:

$$y = \operatorname{bg} \tan x \Leftrightarrow \tan y = x, y \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$y = \operatorname{bg} \tan x$
Nu is $x \in \mathbb{R}$ en $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$



Kansellatie vergelijkingen

$$1) \tan(\operatorname{bg} \tan x) = x \text{ as } x \in \mathbb{R}$$

$$2) \operatorname{bg} \tan(\tan x) = x \text{ as } x \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

Bepaal:

$$\textcircled{1} \tan(\text{bg tan } \overset{\mathbb{R}}{\textcircled{10}}) = x$$

$$\mathbb{R} = D_{\text{bg tan}}$$

$$\Rightarrow x = 10 \quad (x = 10 \in \mathbb{R})$$

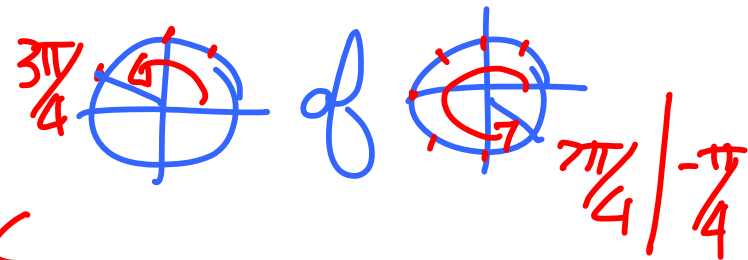
$$\therefore \tan(\text{bg tan } 10) = 10$$

\textcircled{2} Bepaal zonder in zak rekenaar:

$$\tan x = -1 \quad (\text{baie antwoorde})$$

verwysingshoek is $\pi/4$

Teken:



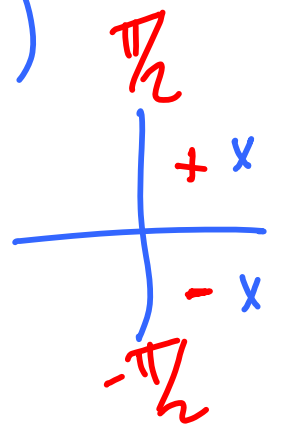
$$\text{Dan } x = 3\pi/4 + \pi k \quad \text{of} \quad x = -\pi/4 + \pi k$$

$$k \in \mathbb{Z}$$

③ $y = \arctan(-1)$ (een unieke antw.)

$\Leftrightarrow \tan y = -1$ en $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

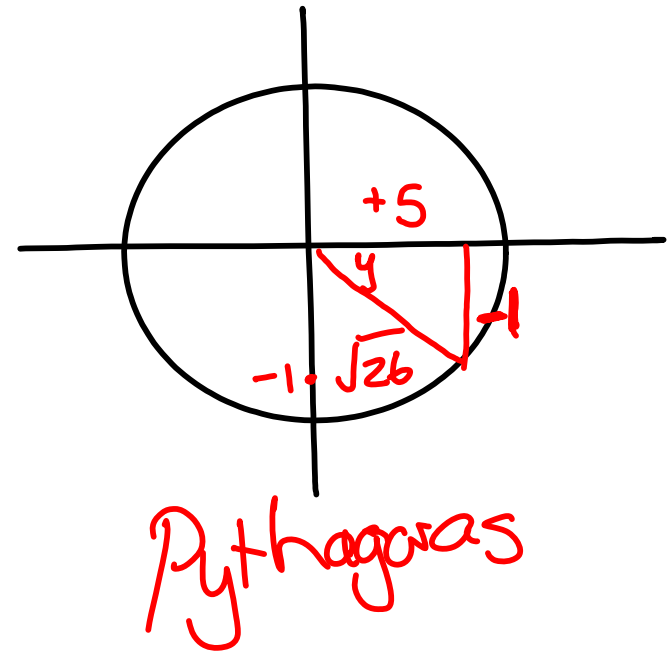
$\Rightarrow y = -\frac{\pi}{4}$ (Nie $y = \frac{\pi}{4}$)



④ $\cos(\arctan(-\frac{1}{5}))$

$x = \cos y$
 waar $\arctan(-\frac{1}{5}) = y$
 $\Leftrightarrow \tan y = -\frac{1}{5}$ en $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$
 ($= \frac{1}{5}$)

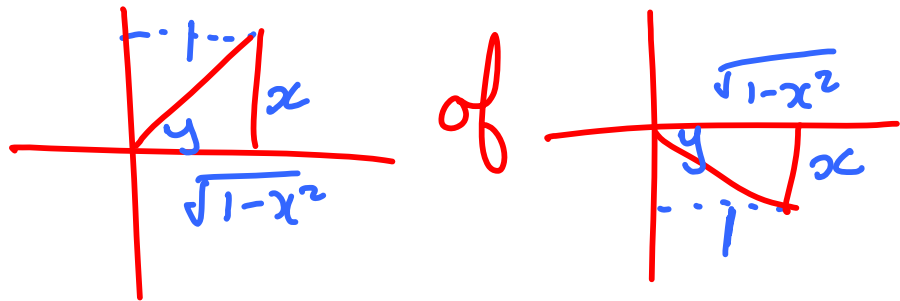
Dus $\cos y = \frac{5}{\sqrt{26}}$



⑤ $\tan(\arcsin 0.5) = x$?
 Laat $\arcsin \frac{1}{2} = y \Leftrightarrow \sin y = \frac{1}{2}, y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
 $\Rightarrow y = \frac{\pi}{6}$

Dan $\tan y = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$

⑥ Bewys dat $\cos(\sin^{-1} x) = \sqrt{1-x^2}$
 $y = \sin^{-1} x \Leftrightarrow \sin y = \underline{\underline{x}}$ en $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$



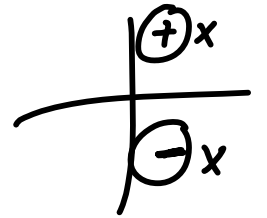
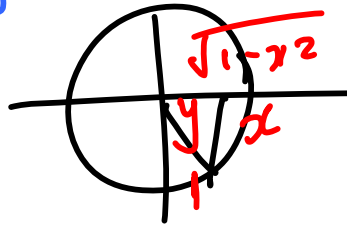
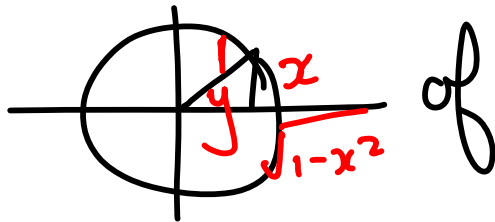
Dan $\cos(\sin^{-1} x) = \cos y = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$

7

Vereenvoudig

$$\tan(\sin^{-1}x)$$

Laat $\sin^{-1}x = y \Leftrightarrow \sin y = \underline{x}$, $y \in \underline{[-\frac{\pi}{2}, \frac{\pi}{2}]}$



Dus $\tan y = \frac{x}{\sqrt{1-x^2}}$

8

Bepaal D_g en W_g as $g(x) = \sin^{-1}(3x+1)$

Definisievers. van $y = \sin^{-1}(x)$:

$y = \sin^{-1}(x) \Leftrightarrow \sin y = x$ en $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ en $x \in [-1, 1]$

$\Rightarrow -1 \leq x \leq 1$

Definisievers. = ?

Now $g(x) = \sin^{-1}(3x+1)$

Plan: $-1 \leq 3x+1 \leq 1$
 $\Rightarrow -2 \leq 3x \leq 0 \Rightarrow -\frac{2}{3} \leq x \leq 0$

$$D_g = \left[-\frac{2}{3}, 0\right]$$

$$* W_g = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

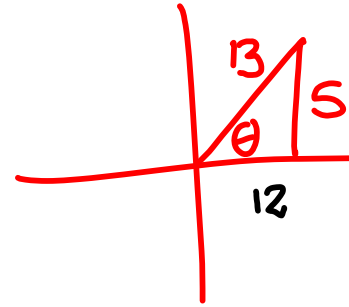
(uit definitie van
 $y = \sin^{-1}x$)

MaaK die beste gebruik van elke
geleentheid = Doen die rechte ding
op die rechte tyd.

$$\times \cos(2 \sin^{-1}(\frac{5}{13}))$$

$$\text{Laat } \theta = \sin^{-1}(\frac{5}{13})$$

$$\Leftrightarrow \sin \theta = \frac{+5}{13} \text{ en } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$



$$\begin{aligned} \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= \left(\frac{12}{13}\right)^2 - \left(\frac{5}{13}\right)^2 \\ &= \frac{119}{169} \end{aligned}$$

$$\begin{aligned} \sin 2\theta &= 2 \sin \theta \cos \theta \end{aligned}$$

Los x qd:

$$\ln(3x-2) + \ln(x+1) \leq 0$$